WEEK 3 · LECTURE L2 · UNIT I

Business Intelligence Fundamentals

From source systems to a single source of truth: warehouses, metrics, dashboards and data quality, and why so many organizations have no single source of truth.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Explain the layers of a BI stack: sources, integration (ETL), warehouse and marts, metric layer, dashboard.
2. Explain why a metric is an agreement, and specify the six questions that define one.
3. Apply Ackoff, Inmon and Kimball, Goodhart and Campbell, Wang and Strong, and Chen, Chiang and Storey to HR data.
4. Evaluate why a single source of truth is a political as much as a technical achievement, and place an HR function on the BI maturity levels.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 1 and CLO 4.

COMMIT BEFORE THEORY

Commit to an answer first

Payroll reports 8,140 employees, the HR system 8,020 and finance 7,960. What is the root problem?

A

One system has a bug

B

There is no agreed definition and no single integrated source: each system counts differently

C

Finance is always wrong

D

Headcount cannot be measured

EXAM LENS

The answer is B. Often each number is right by its own definition.

THE BI STACK

Layers between a transaction and a decision

Layer	What it does	Common failure in HR
Sources	Systems where work happens: HRIS, payroll, ATS, LMS	Each built for today’s transaction, not history
Integration (ETL)	Extracts, cleans, conforms and loads data; resolves conflicts between sources	Data rekeyed by hand into spreadsheets
Warehouse and marts	Keeps integrated history in one place, organized for questions; a mart is a subject slice, such as an HR mart	Absent: last March is overwritten
Metric layer	One agreed definition per measure, with an owner	Each team defines its own
Dashboards and reports	What decision makers see	Bought before the layers beneath it exist

EXAM LENS

Errors upstream flow downstream. A dashboard bought before integration, warehouse and metric layer exist displays disagreement faster.

THEORY IN DEPTH · INMON (1992; 2005); KIMBALL (1996; 2013)

What a warehouse is, and two ways to build one

Inmon’s definition

A subject oriented, integrated, time variant and non volatile collection of data supporting management decisions. Build one integrated enterprise warehouse first, then departmental marts: slower, more consistent.

Kimball’s dimensional model

Facts (measurable events: an exit, a hire) linked to dimensions (who, where, when, which role) in a star schema. Build business focused marts with shared dimensions: faster, but needs discipline on shared definitions.

Why it matters for HR

Operational systems run the business and hold the present; the warehouse lets you analyze it and holds integrated history. An HR fact table of appraisal and attrition events joined to an employee dimension is a Kimball design.

EXAM LENS

Compare the two approaches on speed, consistency and governance; neither is simply better.

A METRIC IS AN AGREEMENT, NOT A FORMULA

Six questions that turn “attrition rate” into a definition

Question	Choices to settle
Numerator	All leavers or voluntary only? Retirements? End of fixed term?
Denominator	Opening, closing or average headcount? Contractors in or out?
Period	Financial year or rolling twelve months? Annualized from a quarter?
Grain	Whole firm, department, site or manager? At what size do you suppress?
Owner	Who may change the definition, and who must be told?
Restatement	When it changes, is history recalculated or left as published?

EXAM LENS

Two teams using different answers will report different numbers, and both will be “right”. Name the six questions in any answer on HR metrics.

Course framing.

THE SINGLE SOURCE OF TRUTH

Why so many organizations have none

What it means

Each fact has one authoritative home (the system of record, Week 4) and one definition (the metric layer), with data combined on common keys such as employee ID, and an owner who decides and records changes.

Two systems of record

An old on premise HRIS and a new cloud HRIS both holding employee records produce conflicting truths.

The spreadsheet paradox

Universal use of private spreadsheets with low satisfaction is not success; it is the fallback when no system is trusted.

EXAM LENS

Agreeing one definition of an employee means somebody’s version of the truth must lose. That makes it a political achievement. Without it, meetings argue about whose number is right instead of what to do.

RECENT EVIDENCE · DEHGHANI (2022)

Data mesh is the main current alternative to one central warehouse: each business domain owns its data as a product, under shared standards. A single source of truth can be a shared agreement rather than a single store.

THEORY IN DEPTH · ACKOFF (1989)

From data to wisdom

The core idea

Data are symbols; information answers who, what, where; knowledge answers how; wisdom judges what is worth doing.

How it works

Each level adds context and purpose. A dashboard sits at information; a decision needs knowledge.

Limits and critique

The neat pyramid is criticized: in practice the levels blur, and judgment shapes even raw data.

IN PRACTICE

An organization can hold vast data in spreadsheets but little information, because nobody has agreed what the numbers mean.

EXAM LENS

Use the hierarchy to explain why more data does not automatically mean better decisions, then note the critique.

THEORY IN DEPTH · WANG AND STRONG (1996)

Data quality is fitness for use

Intrinsic

Accuracy, objectivity, believability, reputation

Contextual

Relevance, timeliness, completeness, appropriate amount

Representational

Interpretable, easy to understand, consistent

Accessibility

Can be accessed, and is secure

Real HR data is never clean

A course marked “not started” that still carries an assessment score; age and tenure that cannot both be true; system names in one table that match nothing in another. Write down every cleaning rule you apply: a figure nobody can reproduce is an opinion.

Practitioner checklist

Accuracy, completeness, consistency, timeliness, validity and uniqueness.

EXAM LENS

Classify each data problem into a family: contradictory records fail intrinsic quality; inconsistent labels fail representational quality. The dimensions overlap, and fitness depends on the task.

THEORY IN DEPTH · GOODHART (1975); CAMPBELL (1976)

A measure that becomes a target stops measuring

The core idea

When people are rewarded on a number, they change their behavior to move the number.

How it works

Campbell: the more a quantitative indicator drives social decisions, the more it is corrupted and corrupts the process it monitors.

Limits and critique

Not every target fails; pairing a metric with a counter metric limits gaming.

IN PRACTICE

Reward recruiters on time to hire and quality of hire may fall; an HR team rewarded for chatbot deflection will deflect, not resolve.

EXAM LENS

Pair Goodhart with a counter metric in any answer that proposes HR KPIs.

A DASHBOARD IS A DECISION SURFACE

What makes a dashboard useful

Every tile needs an owner and an action

If nobody is accountable and nothing follows from the number moving, the tile is decoration.

Comparison beats level

A bare number cannot be interpreted; meaning comes from comparison with last period, a target or a peer unit.

A watched number gets managed

Publish time to fill, and time to fill will improve, possibly at the cost of quality of hire.

CHEN, CHIANG AND STOREY (2012)

Three generations of business intelligence and analytics

BI and analytics 1.0

Structured data in warehouses; reporting and dashboards.

BI and analytics 2.0

Web and unstructured data: text, clicks, social media.

BI and analytics 3.0

Mobile and sensor data; real time, context aware analysis.

EXAM LENS

Most HR analytics still works at 1.0, on structured transactional data (Course framing).

BI MATURITY IN HR

Five levels, and what makes BI succeed

Five levels

1 Data collection (manual extracts) · 2 Reporting (standard descriptive reports) · 3 Analysis (diagnostic, segmentation) · 4 Predictive analytics · 5 Prescriptive analytics (optimization, simulation).

Success factors

Data quality, analytical capability, a culture willing to act on uncomfortable findings, integration with decision processes, and stakeholder trust.

RECENT EVIDENCE · MARGHERITA (2022)

A systematic review maps 106 HR analytics research topics into three areas: enablers (technological and organizational), applications (from descriptive to prescriptive) and value (for employees and the organization).

Course’s Unit 2 teaching note.

APPLY

Measuring time to hire honestly

Teaching case (Course framing)

A Noida software firm reports an average time to hire of 32 days. Recruiters start the clock when a job is posted; the HR system starts it when the requisition is approved; payroll records the joining date. Senior roles take 90 days, campus hires 15. The CEO wants one honest number.

Work it through

1. Which start and end points would you define, and why?
2. Which systems and keys would you integrate?
3. Is an average the right measure? What else would you show?
4. Who should own the definition?

GUIDANCE

Start: approved, advertised or opened? Stop: offer made, accepted, or first day? Split by role, location and channel; one extreme vacancy moves a mean but not a median; report the quartiles; flag requisitions still open, or the slowest cases vanish.

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
A dashboard is the BI system	It is the visible end of a pipeline of sources, integration, warehouse and definitions
Data quality means accuracy	Wang and Strong: also relevance, timeliness, interpretability and access
Different numbers mean someone is wrong	Often each is right by a different definition
More data means better decisions	Ackoff: data become useful only with context and purpose

COMPARE AND EVALUATE

Six lenses on turning data into decisions

Theory	Use it to	Main weakness
Ackoff (1989)	Explain why data are not decisions	The pyramid oversimplifies
Inmon	Build one consistent enterprise warehouse	Slow
Kimball	Build fast business marts	Needs discipline on shared dimensions
Wang and Strong (1996)	Classify data problems	Overlapping dimensions
Goodhart; Campbell	Anticipate gaming of KPIs	Not every target fails
Chen, Chiang and Storey (2012)	Place analytics by generation	A technology lens, not an HR one *

* Course framing.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (1 of 2)

An Indian hospital chain has three different headcount figures. Using the BI pipeline, Inmon’s definition of a warehouse and Wang and Strong’s data quality categories, explain the problem and design a single source of truth. **20 marks**

A strong answer: traces the problem to definitions and integration; applies Inmon or Kimball; uses Wang and Strong beyond accuracy; proposes integration, definitions and governance, with a limit.

“A metric is an agreement, not a formula.” Explain this statement and discuss its implications for building HR dashboards. **20 marks**

A strong answer: covers the six defining questions; links to the metric layer and single source of truth; applies Goodhart and Campbell; proposes counter metrics.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (2 of 2)

Compare the Inmon and Kimball approaches to data warehousing and evaluate which suits an HR function beginning its analytics journey. **20 marks**

A strong answer: explains both; compares on speed, consistency and governance; reaches a justified, conditional recommendation.

Explain Wang and Strong’s four families of data quality with an HR example of each. **10 marks**

A strong answer: names all four families with a correct, distinct example for each.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **A warehouse keeps history on purpose.**
   Operational systems run the business; the warehouse lets you analyze it.

2. **A metric is an agreement with an owner.**
   Not a formula in a cell: six questions define it.

3. **A single source of truth is political.**
   Someone’s version of the truth must lose.

4. **A dashboard should change a decision.**
   If it changes nothing, it is decoration; if it is gamed, add a counter metric.

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Johnson, R. D., Carlson, K. D., & Kavanagh, M. J. (2025), Chapter 11; Chen, Chiang and Storey (2012).
- Angrave, D., et al. (2016), on why HR analytics capability is weak.

> **NOTE**
> The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Ackoff, R. L. (1989). From data to wisdom. *Journal of Applied Systems Analysis, 16*, 3 to 9.
- Campbell, D. T. (1976). *Assessing the impact of planned social change* (Occasional Paper Series No. 8). Public Affairs Center, Dartmouth College.
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- Dehghani, Z. (2022). *Data mesh: Delivering data driven value at scale*. O'Reilly Media.
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- Margherita, A. (2022). Human resources analytics: A systematization of research topics and directions for future research. *Human Resource Management Review, 32*(2), 100795. https://doi.org/10.1016/j.hrmr.2020.100795
- Wang, R. Y., & Strong, D. M. (1996). Beyond accuracy: What data quality means to data consumers. *Journal of Management Information Systems, 12*(4), 5 to 33. https://doi.org/10.1080/07421222.1996.11518099